Areal Average and Spatial Interpolation of Precipitation Data in ArcGIS Pro for Toronto on February 25, 2022

Hydraulics and Hydrology

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Date: 2026-03-02

Abstract

Tracking rainfall is a cornerstone of hydrological study. Since rain is measured at specific gauge points rather than falling in an even blanket, we have to use spatial analysis to figure out how that water is distributed across the map. This study looks at Toronto's precipitation on February 25, 2022, comparing two specific methods in ArcGIS Pro: Thiessen Polygons for calculating area-wide averages and Inverse Distance Weighting (IDW) for mapping the flow of rainfall.

We pulled daily rainfall figures from four Environment and Climate Change Canada stations and brought them into a GIS environment. The Thiessen Polygon approach works by carving the city into "zones of influence" based on the nearest gauge. This allows us to calculate a weighted average for the whole city. It's a practical way to get a bird's-eye view of the weather; it does operate on the assumption that rainfall is uniform within each zone, a limitation that becomes more obvious when you only have a handful of gauges to work with.

To get a more fluid picture, we used IDW interpolation. Instead of drawing hard lines like the polygons, IDW predicts rainfall at unmeasured spots by giving more "weight" to nearby stations than those further away. The result is a smooth, continuous map of the storm's footprint.

The data showed differences across the four stations, with totals ranging from 0.6 mm to 4.2 mm. The IDW map highlighted that the heaviest rain was concentrated mainly in the northern and western parts of Toronto.

Both methods prove why spatial interpolation is so vital for urban planning. While Thiessen Polygons are great for a quick, efficient average, IDW is superior for visualizing how a storm moves across the landscape. These insights are what engineers need when designing stormwater drains or planning for the next flood.

1. Introduction

Reliable rainfall data matters for pretty much every part of hydrologic analysis, from predicting floods to planning stormwater systems. In Toronto, rain doesn't fall evenly, and even small differences across the city can affect runoff, sewer performance, and how the watershed reacts. For urban planners and engineers, understanding where the rain falls is just as important as knowing how much falls overall.

In practice, precipitation is measured at individual gauge stations, meaning the data represent specific points rather than a continuous surface. Yet most hydrological applications, such as watershed modelling or municipal planning, require rainfall information across an entire drainage area or city boundary. Because rainfall patterns shift over space, it becomes necessary to apply spatial techniques that can extend point measurements into broader area estimates.

Two commonly applied methods address this need in different ways. The Thiessen Polygon approach divides a region into areas of influence around each monitoring station, allowing a weighted average rainfall to be calculated for the whole study area. In contrast, Inverse Distance Weighting (IDW) assumes that locations closer to a gauge are more similar in rainfall than those farther away, producing a gradual, continuous surface of estimated precipitation. In this analysis, both techniques were implemented using ArcGIS Pro to examine how rainfall was distributed across Toronto on February 25, 2022, providing both a representative city-wide average and a visualization of spatial variability.

2. Methodology

2.1 Data Acquisition

Four precipitation gauges within Toronto were used. Daily precipitation data for February 25, 2022, were obtained from Environment and Climate Change Canada.

Recorded precipitation values were:

Gauge Name	***Precipitation (mm)***
Toronto City Centre	1.4
Toronto City	0.6
Toronto Intl A	4.0
Toronto North York	4.2

Table 1: Toronto Gauge Precipitation Values

These values were added to the attribute table of the gauge shapefile in ArcGIS Pro under a new float field named “Preci”.

2.2 Areal Average Using Thiessen Polygons Conceptual Description

The Thiessen Polygon method divides the study area into regions such that each polygon contains the area closest to a specific rain gauge. Each gauge is assumed to represent rainfall uniformly within its polygon. The weighted areal average precipitation is calculated using:

$$P_{ave} = \frac{\sum Ai\ Pi}{\sum Ai}$$

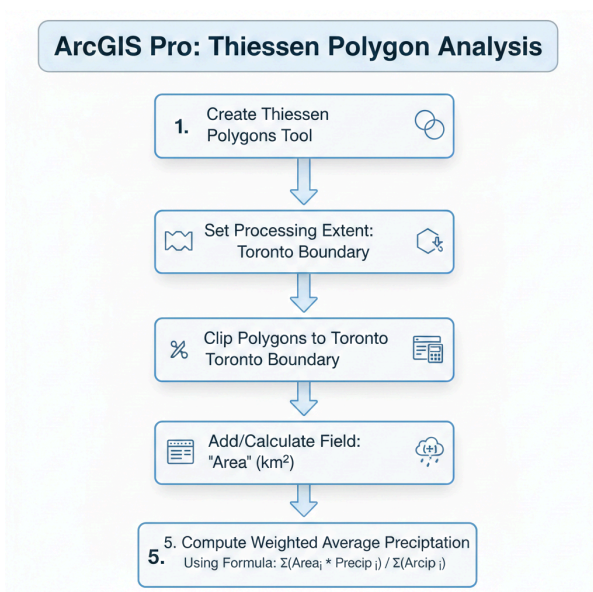
Where

A_i = area of polygon i

P_i = precipitation at gauge i

Procedure in ArcGIS Pro

1. Create Thiessen Polygons using the “Create Thiessen Polygons” tool.
2. Set processing extent to the Toronto boundary.
3. Clip polygons to the Toronto boundary.
4. Add a new field (“Area”) and calculate polygon area (km²).
5. Compute weighted average precipitation using the formula above.



2.3 Interpolation Using IDW Conceptual Description

Inverse Distance Weighting (IDW) estimates precipitation at unsampled locations based on nearby measured values. The influence of each gauge decreases with increasing distance. The general IDW equation is:

$$P(x) = \frac{\sum \frac{Pi}{d_i^P}}{\sum \frac{1}{d_i^P}}$$

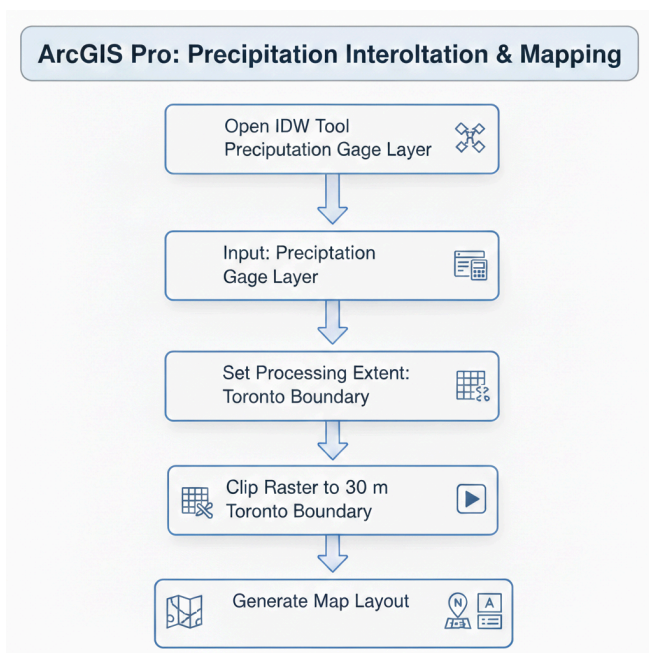
Where:

di = distance to gauge i

P = power parameter

Procedure in ArcGIS Pro

1. Open the IDW tool (Spatial Analyst → Interpolation → IDW).
2. Input: precipitation gauge layer.
3. Z value field: Preci. And Output cell size: 30 m.
4. Set the processing extent to the Toronto boundary and run the tool.
5. Clip raster to Toronto boundary.
6. Generate map layout.



3. Results

3.1 Thiessen Polygon Map

The Thiessen Polygon map splits Toronto into four areas, each tied to one of the precipitation gauges, with each polygon representing the region closest to its station. The highest rainfall was recorded at Toronto North York (4.2 mm) and Toronto Intl A (4.0 mm), indicating that the northern and western parts of the city experienced the most precipitation.

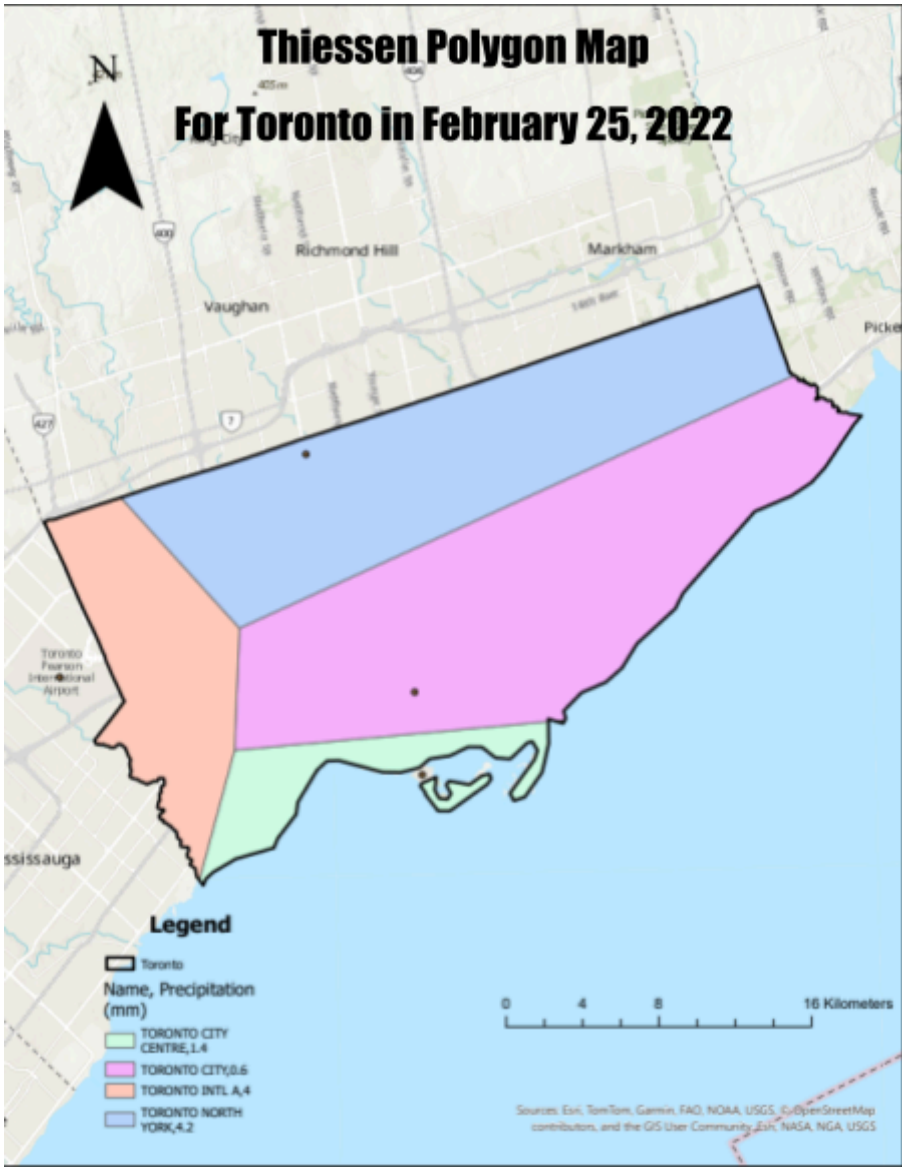


Figure 1: Thiessen Polygon Map

3.2 Areal Average Precipitation

Using calculated polygon areas and recorded precipitation values, the weighted areal average precipitation for Toronto on February 25, 2022, was determined.

Gauge	***Area (km²)***
Polygon 1	101.1
Polygon 2	282
Polygon 3	47
Polygon 4	227
Total ($\sum Ai$)	657.1

Table 2: Toronto Polygon Areas

Multiply Area × Precipitation (From Tables 1 and 2)

$$101.1 \times 1.4 = 141.54$$

$$282 \times 0.6 = 169.20$$

$$47 \times 4.0 = 188.00$$

$$227 \times 4.2 = 953.40$$

Final calculated:

$$P_{ave} = \frac{141.54 + 169.20 + 188.00 + 953.40}{657.1}$$

$$P_{ave} = 2.21 \text{ mm}$$

Areal Average Precipitation = 2.21 mm.

This value represents the spatially weighted precipitation across Toronto.

3.3 IDW Interpolated Map

The IDW interpolation produced a continuous precipitation surface across Toronto. The results show lower precipitation in central Toronto (0.6-1.4 mm), whereas higher precipitation concentrations are in northern and western areas (4.0-4.2 mm). The spatial gradient reflects the distance-weighted influence of higher-rainfall stations.

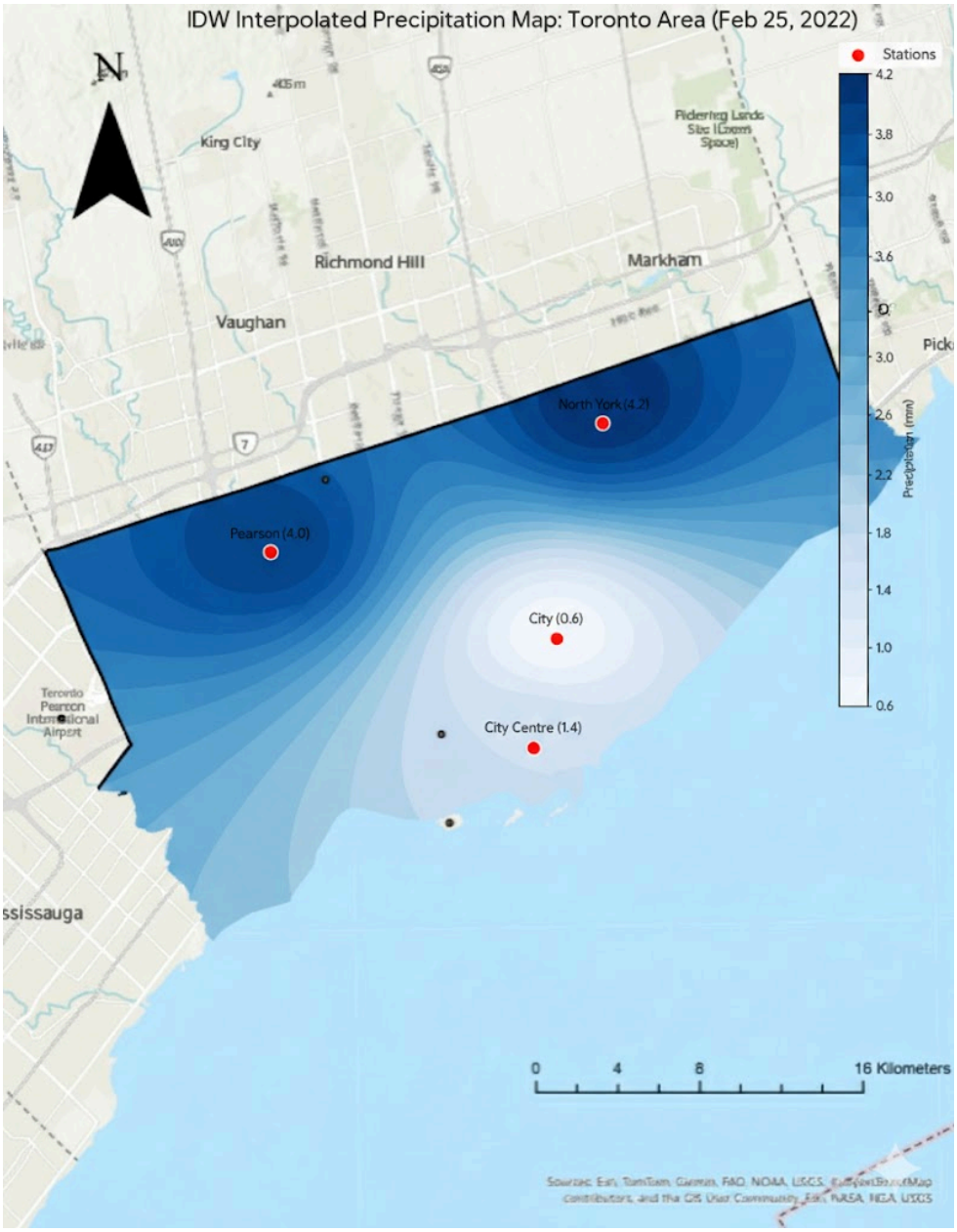


Figure 2: IDW Interpolated Precipitation Map Recreation

4. Discussion

The results indicate that rainfall in Toronto on February 25, 2022, was uneven, with the northern parts of the city receiving noticeably more rain than downtown. The Thiessen Polygon method offers a weighted average based on the area each station covers, which is statistically reliable but assumes rainfall is uniform within each polygon and creates sharp boundaries. By comparison, IDW produces a smoother map that more clearly shows how rainfall gradually changes across the city.

That said, IDW depends heavily on a few key factors. The number of rain gauges matters a lot. In this case, there were only four stations, which limits how detailed the final map can be. The power setting also plays a role because it controls how quickly a station's influence drops off with distance. Changing this value can noticeably shift the rainfall pattern. The placement of gauges matters too. If stations are clustered in one area, that area can end up overrepresented, while places farther away may not be estimated as accurately.

There were a few obvious limitations in this analysis. Since there were only a handful of rainfall stations, both the interpolation and the area-based averaging can only be so accurate. Edge effects are another issue, especially near the city limits, where the estimates depend entirely on stations located within Toronto. The analysis also assumes that rainfall spreads the same way in every direction, which doesn't always reflect how the weather actually behaves.

From a water management viewpoint, these rainfall differences across Toronto really matter. Areas that get heavier rain put uneven pressure on stormwater systems and drainage networks, which can raise the risk of localized flooding. Having accurate rainfall data is also key for modelling runoff and predicting how watersheds respond. Even the way rain is distributed across the city can affect how often combined sewer overflows happen and how severe they are. In the end, knowing exactly where and how much rain falls is crucial for building resilient and sustainable urban water systems.

5. Conclusion

This study used Thiessen polygons and IDW interpolation to examine rainfall across Toronto on February 25, 2022, revealing clear variations in precipitation throughout the city. Thiessen polygons offered a weighted average of overall rainfall, while IDW produced a smoother map showing how rainfall changed across different areas, making the two methods complementary for hydrological analysis. Despite the limited number of gauges, both approaches still provide valuable insights for urban water management and hydrological modelling.